J.D. Miller
Trips to South Africa
Summary
1984 – 2012

March 1984	MINTEK 50, International Conference on Recent Advances in Mineral Science and Technology, Johannesburg.
August 1993	Minerals Engineering 93, Cape Town.
June/July 1997	NSF International Workshop on Multiphase Systems in Mineral and Chemical Processing, Johannesburg/Cape Town.
August 1999	Mineral Processing 99, Cape Town and Short Course, University of Pretoria.
June 2003	Three Seminars, University of Pretoria.
September 2003	XXII International Mineral Processing Congress, Cape Town.
August 2004	Seminar, University of Pretoria.
November 2004	Precious Metals ’04, Cape Town.
April 2007	Invited Seminars, University of Pretoria and receipt of Honorary Doctorate Degree (Doctor Honoris Causa), University of Pretoria.
November 2012	Process Mineralogy 12 (2nd International Symposium on Process Mineralogy, and Minerals to Metals Student Research Day, Cape Town, South Africa.

J.D. Miller
Trips to South Africa
1984 - 2012

MINTEK 50, International Conference on Recent Advances in Mineral Science and Technology, Johannesburg, South Africa, March 6, 1984.

"Fundamental Aspects of Soluble Salt Flotation," Minerals Engineering 93, Cape Town, South Africa, 23–27 August 1993.

"Structures and Surfaces of Multiphase Systems," J. D. Miller, NSF International Workshop on Multiphase Systems in Mineral and Chemical Processing, Johannesburg and Cape Town, South Africa, 27 June–3 July 1997 (organized by P. King and J. D. Miller with assistance from the S.A. Institute of Mining & Metallurgy/U.S. participants supported by NSF). *

"Improved Phosphate Flotation with Nonionic Polymers," Y. Lu, N. Liu, X. Wang, and J. D. Miller (presenter), Mineral Processing 99, Cape Town, South Africa, 5–6 August 1999.

Invited Speaker, "Flotation Technology," J. D. Miller, Short Course, University of Pretoria, Pretoria, South Africa, 2 August 1999.

Three Seminars by J. D. Miller presented at University of Pretoria, Pretoria, South Africa, 14–26 June 2003. (1) "3D Analysis of Particulate Systems by Cone Beam X–ray MicroCT." (2) "Surface Chemistry Considerations in Particle Separations by Flotation." (3) "Thiocarbonate Collectors for Pyrite Flotation."

"3D Analysis of Particulates in Mineral Processing Systems by Cone Beam X-ray Microtomography," J. D. Miller and C. L. Lin, XXII International Mineral Processing Congress, Cape Town, South Africa, 28 September–3 October 2003. **Plenary Session Chairman**.

"Some Thoughts on Sulfide Flotation Chemistry," J. D. Miller, Seminar, University of Pretoria, Pretoria, South Africa, 6 August 2004. **Keynote Speaker**.

"Flotation Chemistry of Pyrrhotite in the Processing of PGM Ores," J. D. Miller and J. C. Davidtz, Precious Metals '04, Cape Town, South Africa, 10–12 November 2004.

Invited Seminars: University of Pretoria, Pretoria, South Africa, 24 April 2007. (1) "The 3D Analysis of Multiphase Particles and Packed Particle Beds by Cone Beam X-ray Micro CT for Improved Understanding of Mineral Processing Systems," J. D. Miller. (2) "Molecular Dynamic Simulations for the Analysis of Interfacial Chemistry in Selected Flotation Systems," J. D. Miller.

Honorary Doctorate Degree (Doctor Honoris Causa), University of Pretoria, South Africa, April 2007.

Invited Keynote Speaker: "The Continued Development of X-ray CT as an Effective Tool in Process Mineralogy," J.D. Miller, Process Mineralogy 12, 2nd International Symposium on Process Mineralogy, Cape Town, South Africa, 5–14 November 2012.

Invited Keynote Speaker: "The Continued Development of X-ray CT as an Effective Tool in Process Mineralogy," J.D. Miller, Minerals to Metals Student Research Day, University of Cape Town, Cape Town, South Africa, 12 November 2012

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*It was during this trip that Professor John Davidtz was first met and instruction on diamond digging/hand jigging resulted in recovery of the Mooi River diamond. This recovery occurred on 24 June 1997 at Farm Klein Rooidraal. Ten tons of Mooi River gravel was prepared and sorted from which the high quality Mooi River diamond (1.42 carat mackel) was recovered, cut, and polished in South Africa and presented by JDM to his wife and sweetheart, Patricia on her birthday, 30 August 1997.

JDM Journal

Travel to Cape Town, South Africa

Process Mineralogy 2012 Conference Keynote Speaker

Minerals to Metals Student Research Day Seminars

5 – 14 November 2012

Travel to Cape Town at the invitation of Barry Wills, Editor of Minerals Engineering, and Megan Becker, Professor at the University of Cape Town, was initiated on Monday 5 November 2012. The invitation was to serve as Keynote Speaker for the Process Mineralogy 12 Conference and to speak at the Minerals to Metals Student Research Day on Monday 12 November 2012. Unfortunately, Granny was not able to join me on this trip because of her recovery from cataract surgery. This excursion made me think about previous trips to South Africa.

The first was in 1984 for the MINTEK 50 celebration. The trip in 1984 included a wonderful expedition to Kruger National Park. Hosts included Roger Paul, Chris Fleming, Rob Dunne and Ian Currans all of MINTEK.

Another memorable visit was the trip on 1997 for the NSF International Workshop on Multiphase Systems in Mineral and Chemical Processing, Johannesburg and Cape Town in June/July. It was during this 1997 trip that Professor John Davidtz was first met and John's instruction on diamond digging/hand jigging resulted in my recovery of the Mooi River diamond. What a thrill! Praise the Lord! This recovery occurred on 24 June 1997 at Farm Klein Rooidraal, the same day that Brooke Hofmann was born in Salt Lake City, Utah. Ten tons of Mooi River gravel was prepared and sorted from which the high quality Mooi River diamond (1.42 carat mackel) was recovered, cut, and polished in South Africa and then presented to my wife and sweetheart, Patricia on her birthday, 30 August 1997. What a great experience and, in addition, the visit resulted in the development of a wonderful friendship with John and Jean Davidtz! Almost all subsequent trips to RSA have included visits with John and Jean at their home outside of Potchefstroom. Since then I have enjoyed 7 subsequent visits, on some of which I was accompanied by my faithful companion, Granny.

More recently the visit in 2007 to receive an honorary doctor's degree, Doctor Honoris Causa, from the University of Pretoria was quite memorable. On this occasion I gave the commencement speech and Granny joined in this trip to the RSA. During our time at the University of Pretoria Roelf Sandenbergh was our host and we stayed at his home in Pretoria. After the ceremonies at the University of Pretoria, we went to stay with John and Jean Davidtz at their farm near Potchefstroom. The 2007 visit included a trip to the

Buffels Vlei game preserve. Photos of this trip are appended. So, this trip in 2012 will be my 9th visit to the Republic of South Africa (RSA). See appended information for more details.

Monday – Tuesday, 5 – 6 November 2012

Departure from SLC was on Monday 5 November 2012 through Atlanta to Johannesburg on Delta, economy class. Then from Johannesburg to Cape Town on South African Airways arriving 9 pm Tuesday 2012 local time in Cape Town. There is a nine hour difference in time, so this corresponds to 12 pm SLC time. The trip took 25 hours with good connections at each stop. Passport Control and Customs in Johannesburg was without difficulty. No Entry Visa was required. I had been booked at the Southern Sun - Newlands Hotel and the driver who had been sent to collect me at the airport was waiting for me and delivered me to the Southern Sun - Newlands. The Southern Sun is a hotel chain in the RSA and other countries in Africa. They may have half a dozen hotels in the Cape Town area.

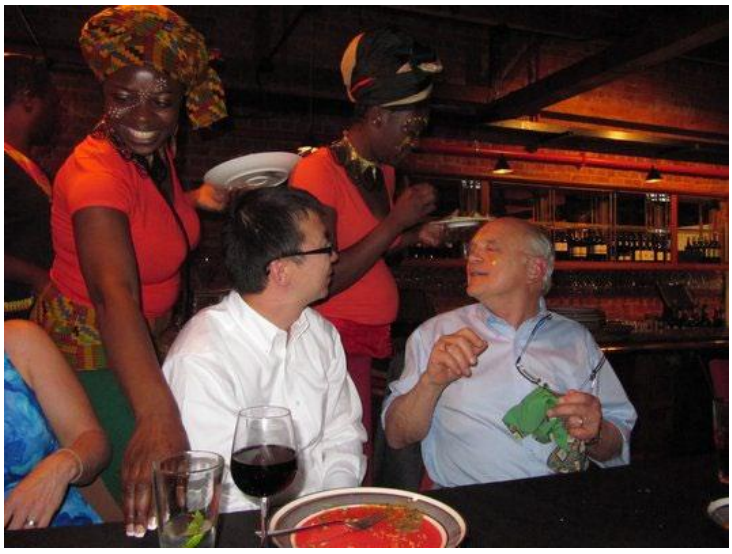
Wednesday – Thursday, 7 – 8 November 2012

The Process Mineralogy Conference, which was held in the Vineyard Conference Center, began Wednesday morning, 7 November 2012. The Vineyard is about a 10 minute walk from the Southern Sun. Wednesday evening I was invited to join S.H. Lau, Rod Browning and their wives (Pei Wah and Christina) for dinner at the Vineyard. We had ostrich, kudo, and fish for dinner. My Keynote presentation was Thursday morning the 8th and was well received with about 150 participants in the conference.
Each morning an excellent breakfast is served at the Southern Sun and at noon we have lunch at the Vineyard Conference Center. The Vineyard is an old estate at the base of Table Mountain that has been restored into a luxury SPA Resort Hotel. The gardens are lovely, the flowers are in bloom (the bougainvillea are beautiful), and ancient tortoises (2 foot diameter shells), who are quite sociable, enjoy the venue. Thankfully I was able to have a swim each day of the conference in the Vineyard pool with the garden and the majestic Table Mountain as a backdrop. Praise the Lord!

Thursday evening we went to a special dinner at the Gold Restaurant with "authentic African tribal entertainment." In addition to service of the food there was dancing and singing, lots of drums. Further the Africans that served us and entertained us also came to paint our faces. I had red and yellow footprints painted on my cheeks and forehead. Of course, there was lots of room on my head for artist creativity. Ostrich and other delicacies were served as part of the meal. See appended menu.



JDM, Dee Bradshaw, and a costumed African at the Gold Restaurant.



Sam Yu, FLSmidth engineer, U of U graduate student, and JDM at the Gold Restaurant.

Friday, 9 November 2012

The conference concluded Friday 9 November 2012 with a reception in the Vineyard garden. I should mention that Saskia Duyvestyn and others, J.P. Franzidis and Cyril O'Connor for example, asked about Patricia. The highlight of the reception was that Dee Bradshaw brought her grandchildren (Jack and a little granddaughter) along with her husband, Michael, to the reception. Dee sends her love and all the best for Granny. Friday night Cyril and Nanette O'Connor took me for dinner at a local restaurant, The Barrister, I believe. We had wonderful dinners of kingklip and steak, and reminisced about previous times we had together. Most memorable was the time we were all together with Janusz and Barbara Laskowski on 9/11 in Istanbul in 2001. Cyril and Nanette send their best to Patricia and would like us to come for a sabbatical at the University of Cape Town (UCT).



S.H. Lau from Xradia, Desh From Mintek, JDM and others at the Process Mineralogy 12 reception.

Saturday, 10 November 2012

The weekend schedule is a tour of Cape Town on Saturday and church with Megan Becker and her family on Sunday. On Monday, I will lecture at a UCT activity and participate in a student research meeting. I expect that J.P.Franzidis will be at the meeting on Monday. I understand that his wife Rose will be off to Australia for their daughter's marriage.



Yousef Ghorbani, Ph.D. graduate student and Mymoena (May moon ah), Administrative Assistant, both from UCT came for me on Saturday morning. We went for a drive along the coast of the Cape Peninsula with Baris Yildirim (student JKMRC) and Dogan Paktunc (CANMET & University of Ottawa). I had acted as external examiner for Yousef's PhD thesis and he desires to join us at the U of U as a Postdoctoral Fellow. We drove along the base of Table Mountain, past Lion's Head on the west side of the peninsula to Camps Bay. At certain seasons whales are cited along the Cape Peninsula coast but I understand it is a bit late and they may have already migrated farther south. There was no shark warning posted and so some people were surfing. The

water is still pretty cold on this side of the Cape Peninsula. Further south we came to Hout's Bay with more surfers and seal island. We saw lots of seals in this area and where there are seals we can expect to find sharks, but none were seen. At Hout's Bay we stopped for coffee and ice cream. From Hout's Bay we continued to Noordhoek and then crossed over the peninsula to the east side, the Indian Ocean side, and came to Kalk Bay Harbour where we stopped for lunch having klingklip and snoek with chips and an Appletiser for drink. Many seagulls were anxiously waiting for us to finish so that they could enjoy the leftovers. We then returned to Cape Town and to the Southern Sun in time for me to do some work for the remainder of the day. The others returned to visit the Boulders beyond Simon's Town where a penguin colony is found. Farther south on the Indian Ocean side you come to Miller's Point.



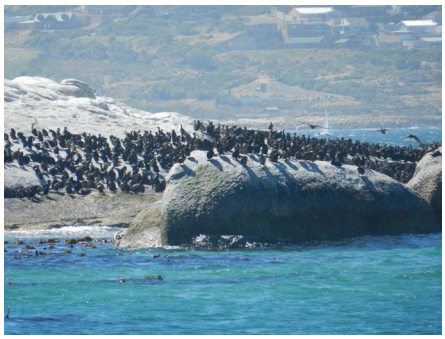
Tour along the South African coast, the Cape Peninsula, with Yousef Ghorbani, Baris Yildirim, and Dogan Paktunc, participants in the Process Mineralogy 12 Conference. Mymoena from UCT took the photo.



Sentinel Peak at Hout Bay.



Sentinel Peak at Hout Bay.



Penguin colony at the Boulders.






Penguin Buddies at the Boulders.

Sunday, 11 November 2012

Joe and Marilyn Coetzee, Megan Becker's parents, came to collect me for Worship Service at Mowbray Baptist Church. We had a good time even though the International translation was used. The message, from Nehemiah Chapter 8, was entitled "What is True Joy?" Prior to the message, Gavin and Megan dedicated their son, Peter, who will be one year old next week. Daughter, Anna, also participated in the dedication. After church we went for lunch at Gavin and Megan's house with their extended family. We had shishkabobs, salads, corn-on-the-cob, etc. with fruit and ice cream for desert. The Coetzee's returned me to the Southern Sun where I did some work and prepared my presentation to be made tomorrow for the "Minerals to Metals Student Research Day" at Belmont Square for the University of Cape Town students.

Monday, 12 November 2012

Professor J-P Franzidis came for me at the Southern Sun - Newlands at 7:30 am Monday morning. We had breakfast and then left for the "Minerals to Metals Student Research Day" at Belmont Square for the University of Cape Town students. I was Keynote Speaker and we began with my presentation, the same as was given at the Process Mineralogy 12 Conference last week at the Vineyard Conference Center. The lecture was well received and we had a good day with student presentations and discussions. Lunch was taken at Belmont Square. One student, Thandazile Moyo, is studying the electrochemistry of the chalcopyrite/ammonia leaching system, a system that Leo Beckstead studied for his PhD thesis about 40 years ago at the University of Utah under

my supervision. Thandazile had recognized our initial research on this system and we had good discussion with a suggestion to do electrochemical AFM experiments. African gifts were given in appreciation for the Keynote Lecture including what appears to be ostrich jerky. J-P then returned me to the Southern Sun in time for evening popcorn. The hotel provides this popcorn treat every evening, most enjoyable.

Tuesday, 13 November 2012

After breakfast, the morning was spent at the Southern Sun – Newlands. A driver came for me at noon to deliver me at the airport for a 2:10 pm departure on a South African Airways flight to Johannesburg. In Johannesburg I got the long Delta flight 201 to Atlanta and then to SLC on Delta 1105. During the return I enjoyed reading "Racketeer," John Grisham's latest novel. For both Delta flights the adjacent seat was vacant and I was quite comfortable although the total travel time was about 24 hours. As always, is great to return home and see my old girlfriend. The trip has been wonderful with the Lord's continued blessings.

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GOLD Restaurant, Cape Town, Spring Menu 2012

CAPE SMOKED SNOEK PATÉ
Recipes for Gestampte Snoek (pounded snoek) were brought to the Cape Colony by Malay slaves in the 17th century. It is similar to “sh cream”, popular amongst British travellers in the 1920’s and 1930’s. Our version has a hint of chilli and a good splash of fresh lemon. Enjoy with the pot bread.
(Vegetarian Substitute – Egyptian butterbean mash and dukkah)

XHOSA POT BREADS
Individual Xhosa pot breads

SOUTH AFRICAN BILTONG DIP
The word biltong is from the Dutch bil ("rump") and tong ("strip" or "tongue"). South Africa’s favorite snack, dried spiced beef. Here we use it in a tasty dip which can be enjoyed on the pot breads.

MALAY ROTI WITH OSTRICH MINCE
Roti wrapped around mildly spiced Ostrich mince served with creamed blatjang (chutney).
(Vegetarian Substitute – Mildly spiced steamed mixed vegetable roti)

TUNISIAN BRIOUATS
Tunisian Briouats are small crispy pastries which are stu-ed with potato and garlic.

ZAMBIAN IMFINO PATTIES
Two of Africa’s most important staples, mealie meal and spinach, are combined to make gorgeous, crispy patties which are gently fried until golden.

SOUTH AFRICAN CHICKEN PIE
Dutch settlers introduced chicken pie to the South African table and they were amongst the earliest bakes sold by the ‘free bakers’ in the colony. This □avourful dish is associated with celebrations Christmas, birthdays, and especially Cape Malay wedding feasts.

NAMIBIAN VENISON POT
The recipe for a good meat pot would have been passed down from mother to daughter. Our mothers have all contributed to this recipe which is made from Organic Springbok (which could not run faster than our chef…) and □avoured with earthly and ground spices.

NORTH AFRICAN COUSCOUS
Couscous is eaten all over North Africa and can be prepared as a meal on its own, with a main course, or as a dessert. To accompany our gorgeous Venison Pot, we have prepared the couscous with corn, herbs and dates.

TANZANIAN MCHICHA WA’NAZI
Spinach (Morog or moroggo) is eaten widely all over Africa in di-erent forms. This is a West African inspiration with the addition of coconut and ground nuts.

MALAY LENTIL DHAL

Lentils are an important part of an Indian meal, and we o-er a nourishing vegetarian dish that comprises of seductively-spiced brown lentils and yogurt mixed with nuggets of roasted butternut and sweet potato.

TABOULEH SURPRISE

This fresh take on traditional Tabouleh is inspired by a recipe in Evita Bezuidenhout's "Evita's Bossie Sikelela" and R1 of every meal sold at GOLD will be donated to the Darling Trust for work in this community and the rest of South Africa."

MOROCCAN ORANGE AND ALMOND CAKE

A beautiful, moist cake served with a dollop of Amarula Crème fraiche.

FRESH FRUIT PLATTERS

If you ask any rural African what they have for dessert, most of them will say they go into the garden and pick some fruit. Here is our selection from our “garden”

All our food is freshly prepared on the day and any food that is left over is delivered daily to the South African Children’s Home which is responsible for around 43 children. See our web site for more details.



JDM delivering the commencement address at the University of Pretoria where he was presented an honorary doctor's degree, Doctor Honoris Causa, April 2007.

Photos from Buffels Vlei Game Preserve









Photos from 2007 South African trip: top left – ostriches; top right – zebras; bottom left – male water-buck; bottom right – female waterbuck.











Top left – gemsbuck; top right – warthog; bottom three photos – lions.









Top left – water buffalo; top right – crested crane; bottom left and right – JDM and Pat with lion cubs.









JDM and Pat at Three Oaks Farm, home of Jean and John Davidtz near Potchefstroom, South Africa.